

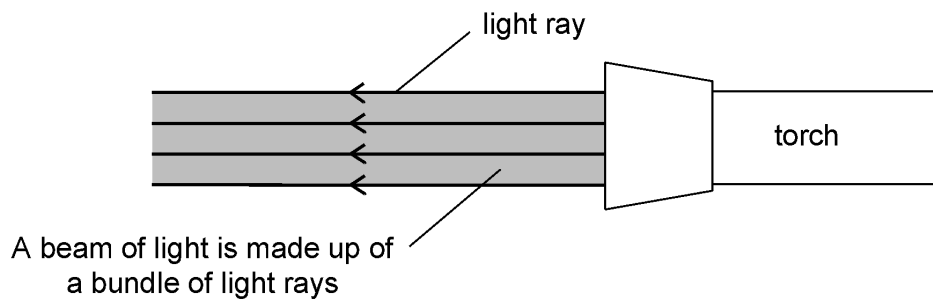
Name : Le Son TungClass/ Reg. No. : S208 (12)Date : 24 August 2026

2026 SECONDARY 2 COMMUNICATION SCIENCE			
S/NO.: 4	TOPIC: Signals	UNIT: Ray Model	NOTES 4

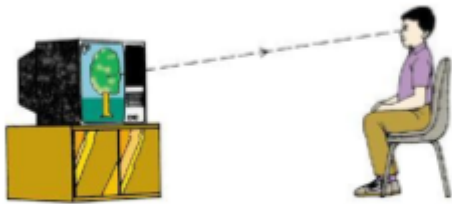
Knowledge	Cognitive Skills or Practical Skills
4.1 Ray model of light	Cognitive Skills (a) Show an understanding that the ray model represents the path taken by light. (b) Recognise that light travels in a straight line, forming shadows when blocked (e.g. eclipse).
4.2 Reflection	(c) Explain how reflection is affected by a smooth and rough surfaces. (d) Describe characteristics of the image formed by a plane mirror. (e) Describe the effects and uses of reflecting surfaces (e.g., plane and curved; include material and design of satellite dishes). (f) Make inferences through observations in everyday life (e.g. as the moon orbits the earth, different parts of it reflects light from the sun, resulting in different moon phases).
4.3 Refraction	(g) Describe effects and consequences of refraction of white light in an optical medium. (h) Show an understanding that the change in the speed of light in different mediums can cause refraction (calculation of angles not required).
4.4 Total Internal Reflection	(k) Recognise conditions for total internal reflection. (l) Describe use of light in telecommunication via optical fiber.
4.5 Dispersion	(i) Describe the dispersion of white light due to refraction by a prism.
4.6 Seeing colours	(j) Explain how we see the color of objects in white light and colored light such as red, blue and green.
4.7 Effects of light	(m) Show an awareness about the impact of light produced by technology, on society and environment (e.g., city lights can improve night visibility but cause light pollution, disorientation of birds and use up a lot of electrical energy).
4.8 Practical Skills	(a) Investigate the characteristics of the image formed by a plane mirror. (b) Investigate that the angle of reflection is equal to the angle of incidence, with respect to the normal. (c) Investigate bending of light.
Essential Questions	1. (F) How does light waves travel? 2. (F) What causes light to bend? F: Factual / C: Conceptual

4.1 Ray Model of Light

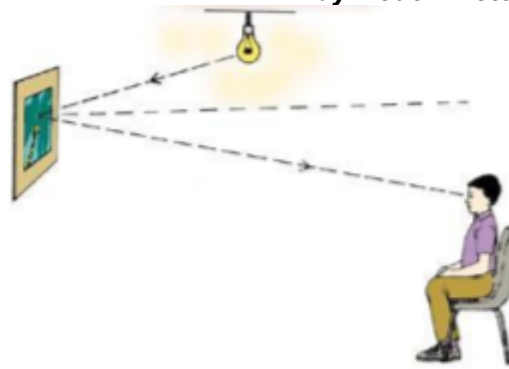
- Light is a form of **energy**
- Light travels in **straight lines** (Ray Model of Light)
- The path along which light energy travels is called a ray.



- Light travels at a speed of **3.0×10^8 m/s** in **vacuum**.
- Light can be bent (reflected or refracted).
- Light is part of the **electromagnetic spectrum**.
- Human eye detects light in a range of colours (V I B G Y O R) called a spectrum.
- This range of colours are mixed to form **white light**
- We are able to see an object only when the light rays coming from it **enter** our eyes.



Luminous objects emit light.
We are able to see them directly.

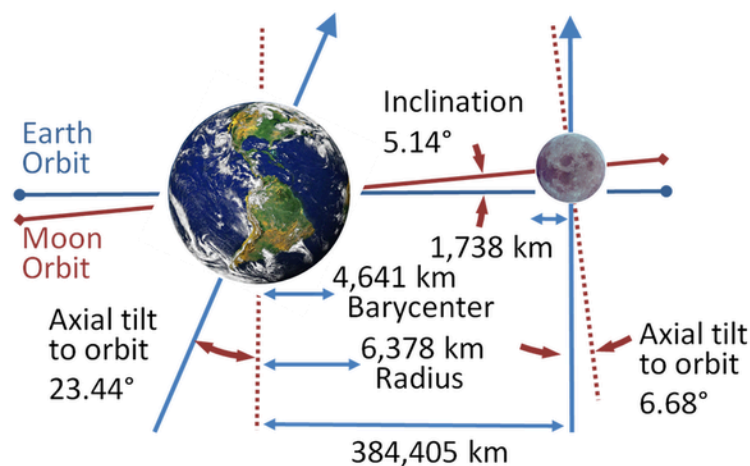


Non-luminous objects do not emit light.
We are able to see them because they **reflect** light from a (light) source.

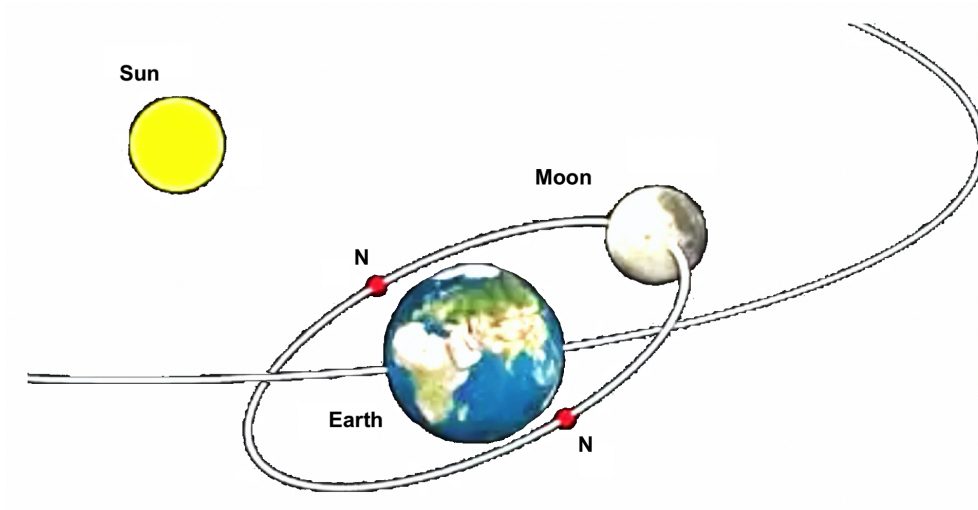
Eclipse

One of the consequences of light travelling in a straight line is the formation of shadows. When the earth casts its shadows on the moon, a lunar eclipse will happen. When the shadows of the moon is casted on earth, a solar eclipse will happen.

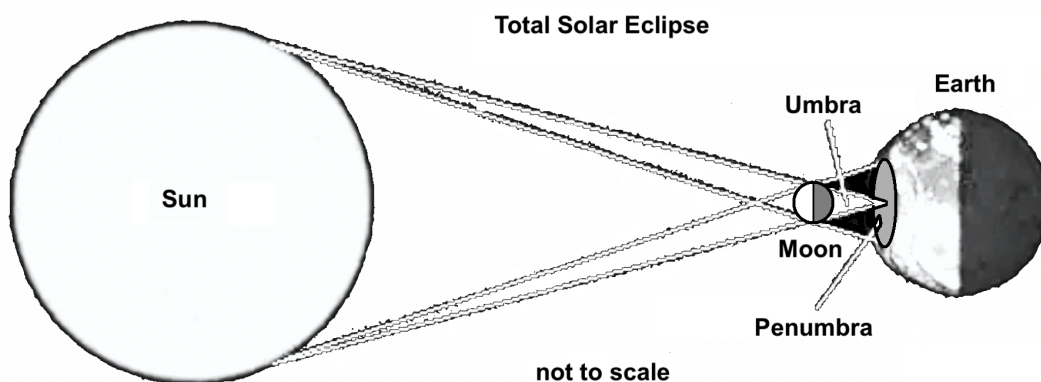
The eclipses do not happen every full moon or new moon because of a reason. There is an inclination of the Moon's orbit with respect to the Earth's orbit around the sun. These orbits are not coplanar, but have an angle of about 5° between them as shown below.



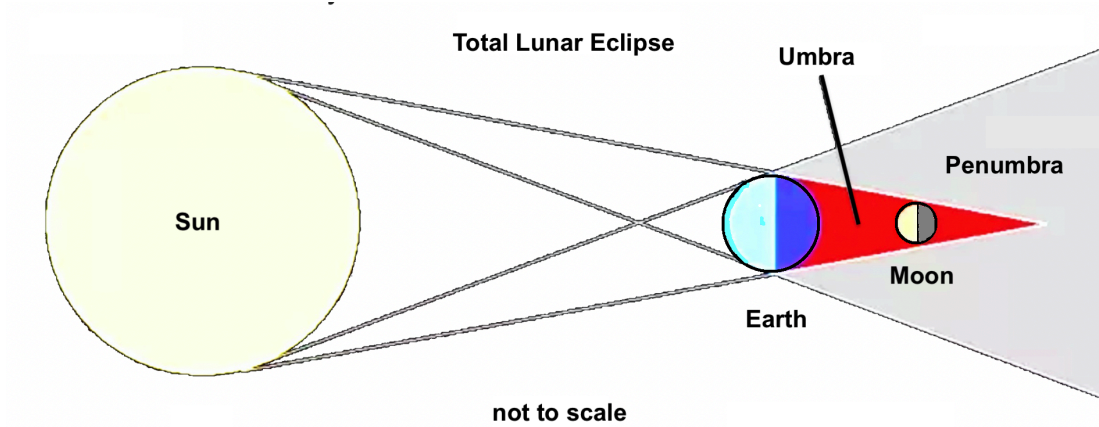
This means the planes of the two orbits intersect in a line. The end points of this line are two points on the Moon's orbit called as nodes. These are shown by the letter N in the image below.



For 2 intersecting circles, there will be 2 sets of nodes. Usually, there will be 2 lunar eclipses and 2 solar eclipses every year. Furthermore, the eclipses will always come in a pair of solar/lunar eclipses that occurs about 6 months apart. Watch this video to understand how it works here: <https://www.youtube.com/watch?v=ljwZMYy930s>. Eclipses occur when the Sun, Earth and Moon are in straight line. During a solar eclipse, the Moon is between the Earth and the Sun. So the shadow of the Moon falls on the Earth. If you are in the darker shadow, called Umbra, a total solar eclipse can be seen. If you are in the lighter shadow, called Penumbra, a partial solar eclipse will be seen. Since the Moon is between the Sun and Earth in a straight line, it is a New Moon. Hence, a solar eclipse always occurs on a new moon day.



Similarly during a lunar eclipse the Earth is between the Sun and the Moon. If Moon is completely within Umbra, a total lunar eclipse will be seen. A partial eclipse is seen when the Moon is partially out of Umbra. If the Moon is completely in Penumbra, it is still an eclipse (called penumbral eclipse). However, this cannot be seen with eyes and other instruments will be needed to detect the penumbra. Since the Earth is between the Sun and Moon in a straight line, it is a Full Moon. Hence, a lunar eclipse always occurs on a full moon day.



You may like to know that the next visible Lunar Eclipse will occur in the southern part of Singapore on the following date:

Date	Time of greatest eclipse	Remarks
08 Nov 2022	1859*	Moon will appear red

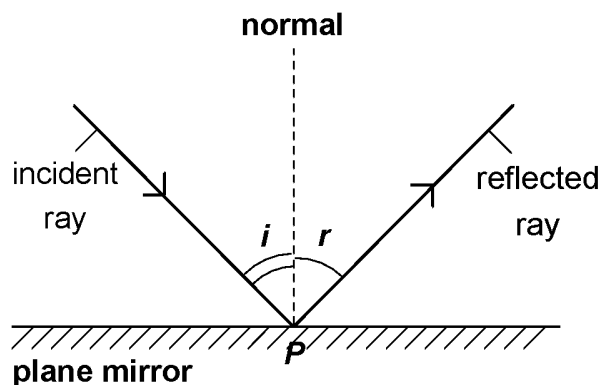
* estimated

(Source: <https://eclipse.gsfc.nasa.gov/SEplot/SEplot2001/SE2019Dec26A.GIF>)

(Source: <https://www.quora.com/What-is-the-geometry-that-leads-to-solar-eclipses>)

4.2 Reflection of Light

- Reflection is light being bounced off after hitting a surface.
- Reflection allows us to see objects in our surroundings.



i : angle of incidence

r : angle of reflection

- Incident ray is the light ray coming towards the surface.

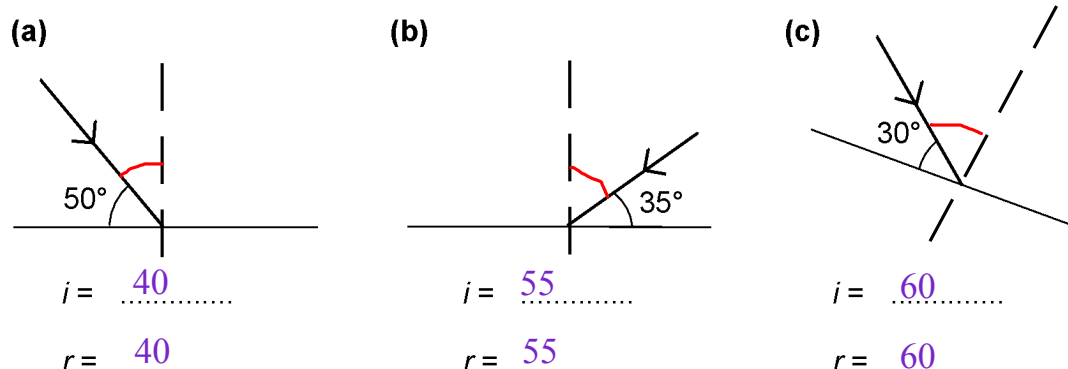
- Reflected ray is the light ray bouncing off the surface.
- **Normal** is an **imaginary straight line** that is (90°) to the
perpendicular surface of the mirror
- **Angle of incidence** (i) is the angle between the and the
angle between the incident ray normal
- **Angle of reflection** (r) is the angle between the and the
angle between the reflected ray normal

Two Laws of Reflection of Light

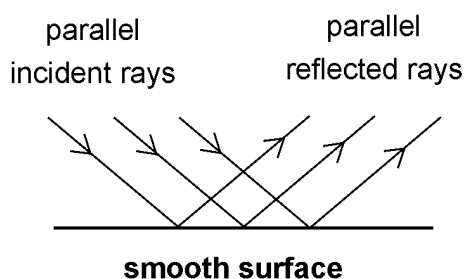
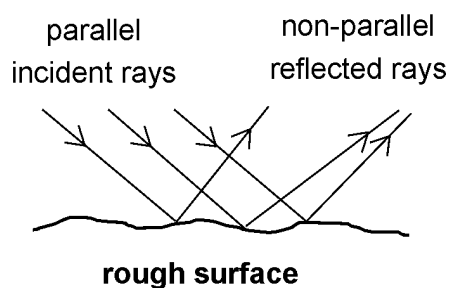
1. The angle of incidence (i) is equal to the angle of reflection (r).
.....
2. At the point of incidence (P), the incident ray, the normal, and the
reflected ray all lie in the same plane.
.....

Example 1

Determine the angle of incidence and angle of reflection and complete the diagram by drawing the reflected ray.

**Types of Reflection – Regular and Irregular**

- All surfaces reflect light.

Regular reflection**Irregular reflection**

Regular reflection	Irregular reflection
- Occurs at smooth surfaces. - Law of reflection applies for each individual ray. - Parallel incident rays are reflected in the same direction (parallel reflected rays).	- Occurs at rough surfaces. - Law of reflection applies for each individual ray. - Parallel incident rays are not reflected in the same direction (non-parallel reflected rays).

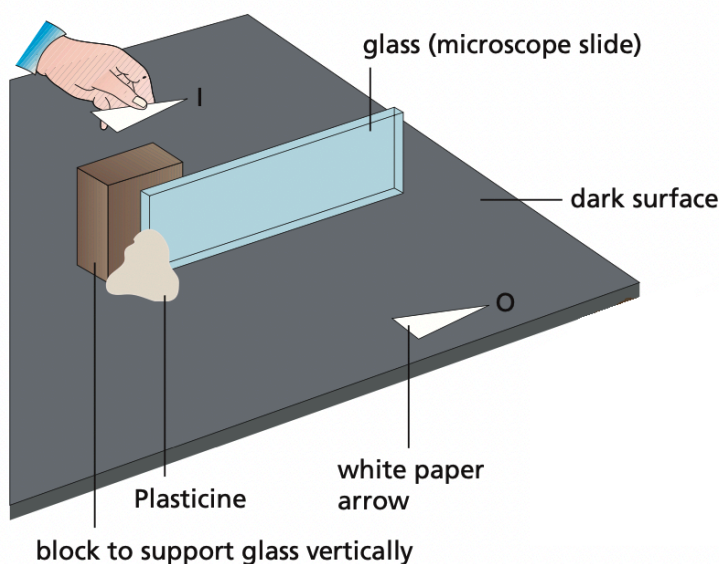
- All the normal at the point of incidence are parallel.

- All the normal at the point of incidence are **not** parallel.

Activity: Characteristics of Image Formed by a Plane Mirror

Support a piece of thin glass on the bench, as in Figure 28.1. It must be *vertical* (at 90° to the bench). Place a small paper arrow, O, about 10 cm from the glass. The glass acts as a poor mirror and an image of O will be seen in it; the darker the bench top, the brighter the image will be.

Lay another identical arrow, I, on the bench behind the glass; move it until it coincides with the image of O. How do the sizes of O and its image compare? Imagine a line joining them. What can you say about it? Measure the distances of the points of O and I from the glass along the line joining them. How do they compare? Try placing O at other distances.



Using the information from the observations above, fill in the blanks.

1. The image is the **same size** and **shape** as the object.
2. It is **upright**
3. It is **virtual**

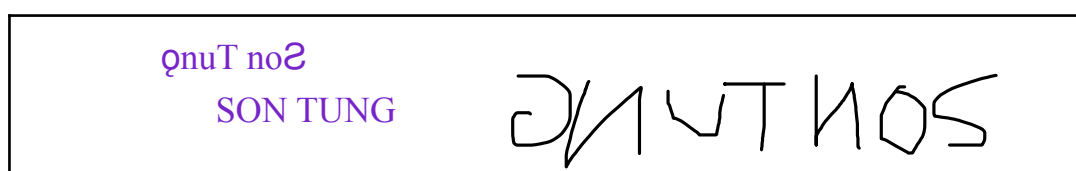
□ **A virtual image cannot be formed on a screen**

□ (Real) Light rays do not meet at the image position.

4. It is formed at the same distance from the mirror, but behind it, as the object is in front of the mirror.

Activity: Viewing words in a Plane Mirror

Write down your name on a piece of paper and hold it facing the mirror. Look at the words in the mirror and write it down below. Describe the words that you see in the mirror.



It is laterally inverted as the letters are flipped right
to left

Drawing Ray Diagram for Plane Mirror

Step 1: Image distance from mirror = object distance from mirror

- ☐ Measure accurately the perpendicular distance between **object O** and the mirror surface. Both distances are measured with respect to the mirror **surface**.
- ☐ Measure the same distance behind the mirror to locate **image I**.

Step 2: Draw light ray (straight line) from the image to the eye

- ☐ Use dashed lines behind the mirror (virtual light rays).
- ☐ Use solid lines in front of the mirror surface (real light rays).

Step 3: Draw incident ray from the object to the point of the reflected rays on mirror surface (point of incidence)

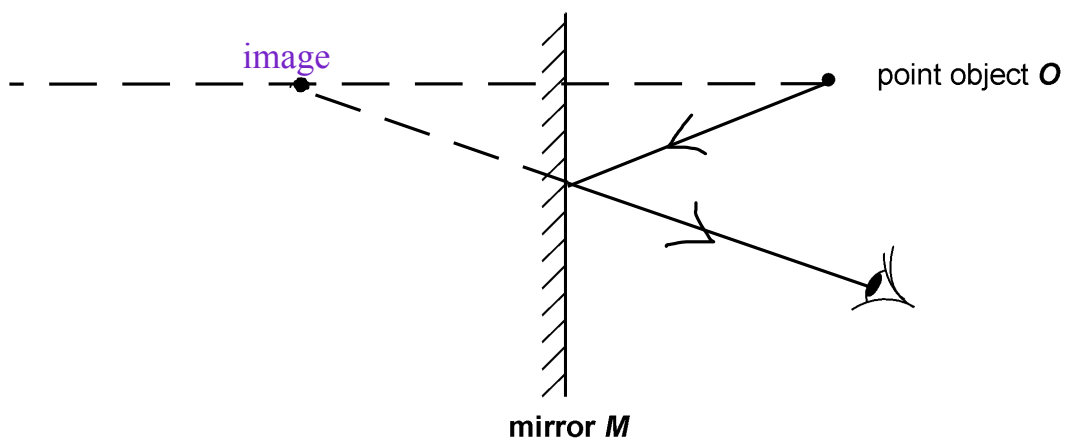
- ☐ *Note:* angle of incidence = angle of reflection

Step 4: Direction of light (object → mirror → observer)

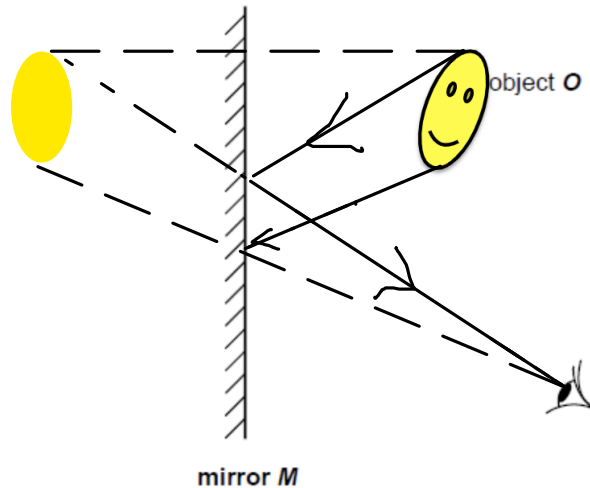
- Draw the arrowheads on the real light rays (solid) lines to indicate direction of the light travelled (real light rays).

Ray Diagram for Plane Mirror

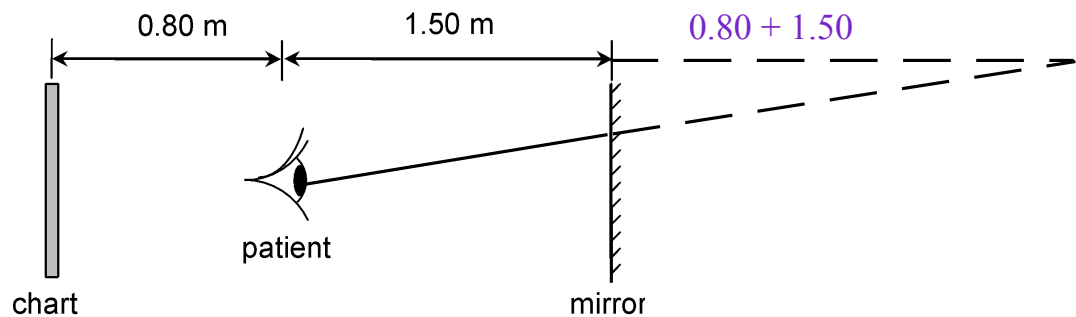
Type 1: Point Object



Type 2: Object

**Example 2**

The diagram shows a patient having an eye checkup. A chart is placed behind her and she sees the image of the chart in the plane mirror.



Determine the horizontal distance between the image of the chart and the patient.

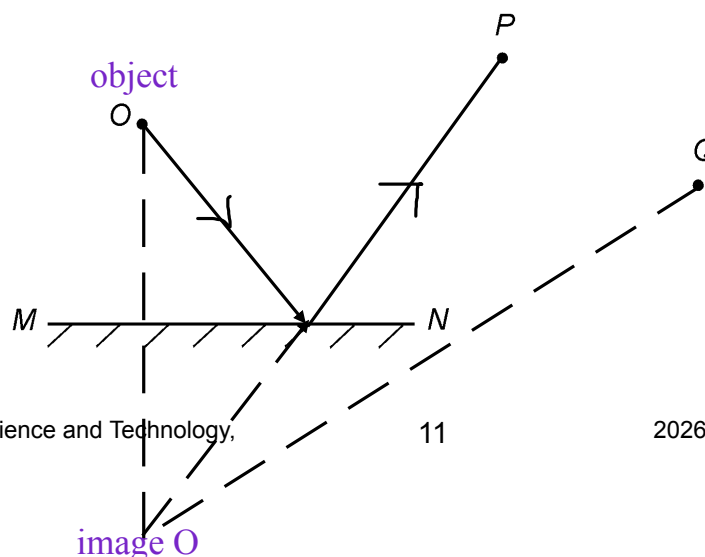
$$1.50 \times 2 + 0.80 = 3.80$$

distance = 3.80m

Example 3

As shown in the diagram, an object O is placed in front of a plane mirror MN.

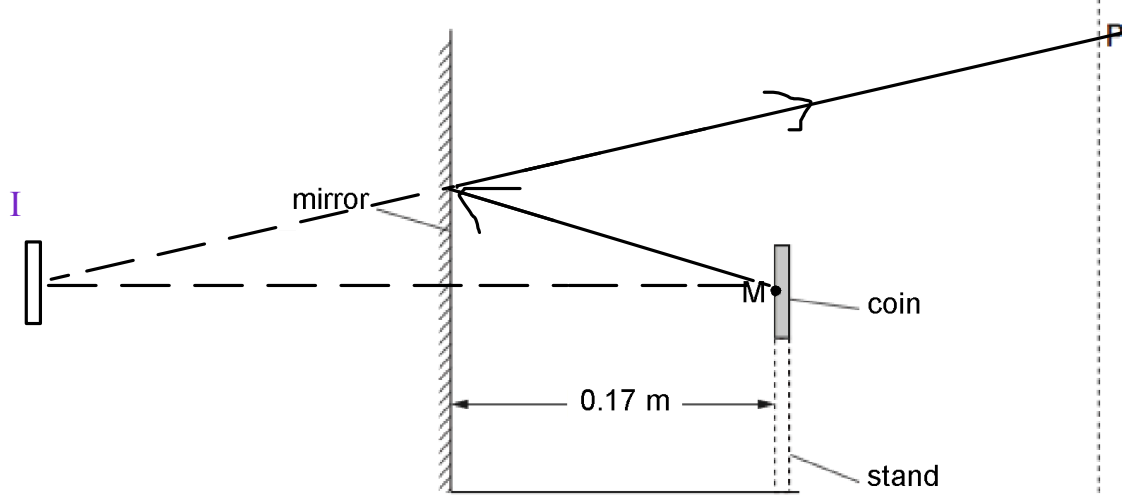
Draw ray diagrams to determine whether the image will be seen if someone is looking at the mirror at point P and Q.



Observer at P *will / will not* see the object.
 Observer at Q *will / will not* see the object.

Example 4

The diagram shows an old coin displayed in a museum. The coin is placed vertically on a transparent stand. A mirror is placed 0.17 m behind the coin to ensure that a visitor standing at P is able to see the back of the coin. M is a point at the back of the coin.



[NOT drawn to scale]

- (a) On the diagram,
- Draw the image of the coin and label it I.
 - Complete the ray diagram to demonstrate how light travels when the person standing at P is looking at the image of point M.
- (b) The visitor is standing 0.15 m away from the coin.
 Calculate the horizontal distance from the person's eye at point P to the coin's image.

$$0.17 \times 2 + 0.15 = 0.49$$

distance = 0.49m

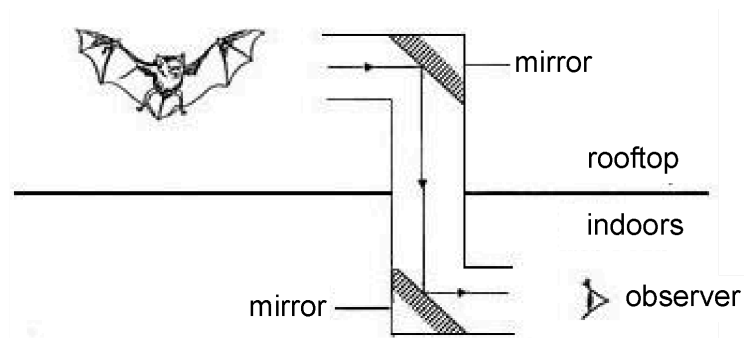
- (c) List down **two** properties of the image.

1. The image is the same size and same shape as the object
2. It is upright

Uses of Reflecting Surfaces

Periscope (plane mirrors)

- In a periscope, two plane mirrors are inclined at 45 degrees
- It helps a person to look over an obstacle such as a high wall.

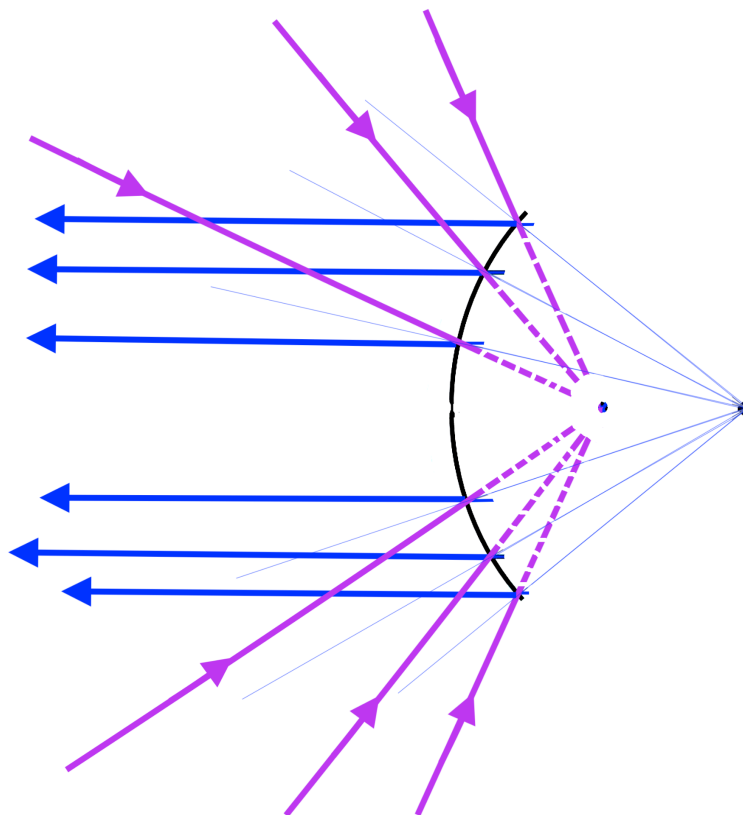


Blind corner mirror (curved mirrors)

- Curved mirrors are fitted at the corners of shops to allow shopkeepers to look out for shoplifters.
- Curved mirrors are used to help drivers to see around blind corners before making a turn.



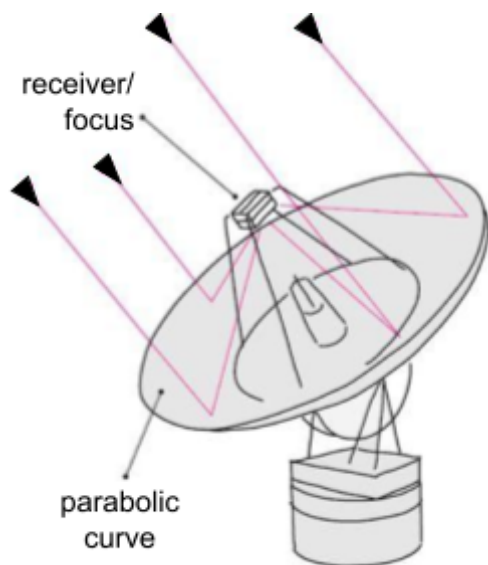
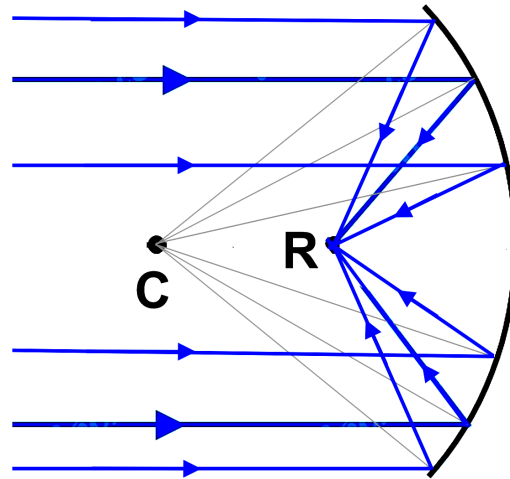
Draw, using ray diagrams, how curved mirrors are used to collect light rays from a large region and focus them into a small area.



The solid lines represent the rays from a wider angle that appears to converge to a point and gets reflected to be parallel to an observer.

Satellite dish (curved surface)

- A satellite dish is usually made of **fiberglass** or **metal** (aluminium).
- It consists of a parabolic (curved, bowl-shaped) surface.
- For a receiving satellite dish, it has a radio receiver located at the focus of the parabola, R.
- The radio rays are reflected from the parabolic surface and concentrated at the focus, R.



Satellite dish**Solar cooker**

- This focusing and amplification property of parabolic reflectors is also used for solar heating and generating solar electricity.

Moon phases

Activity

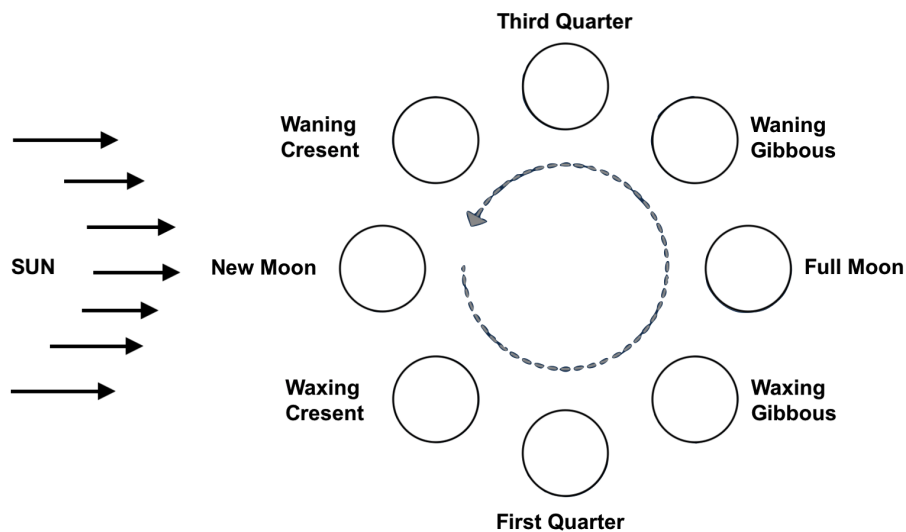
You are given a styrofoam ball with a stick mounted on it to represent the moon.

A bright light bulb acts like a sun.

Your eyes and head will represent an observer on the surface of the earth.

Observe how the moon is illuminated by the sun light as it goes through different days of the month.

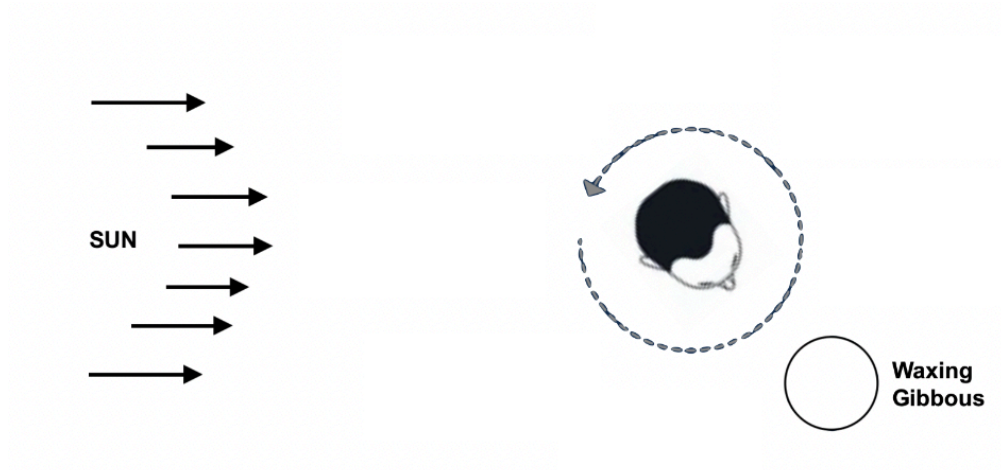
Shade on the diagram below, the different phases of the moon as it moves through the days of the month.

**Observations**

One of consequences of the moon orbiting about the earth is that it will go through the different phases when viewed from earth. The demonstration can be found here:

<https://www.youtube.com/watch?v=wz01pTvuMa0> .

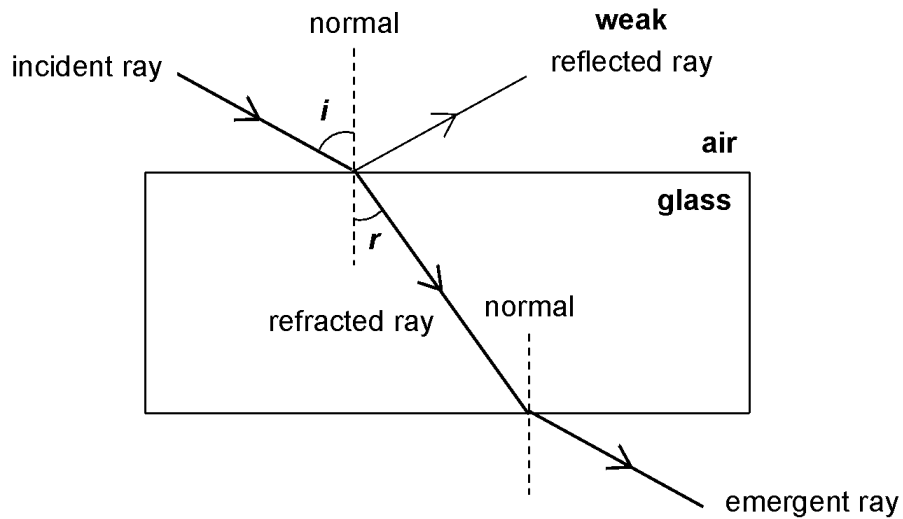
Can you explain how the Waxing Gibbons is created using ray diagrams?



4.3 Refraction of Light

- Refraction is the **bending of light**..... when light travels from one medium to another medium.
- Refraction is caused by a **change in the speed of light**..... as light travels from one medium to another medium.

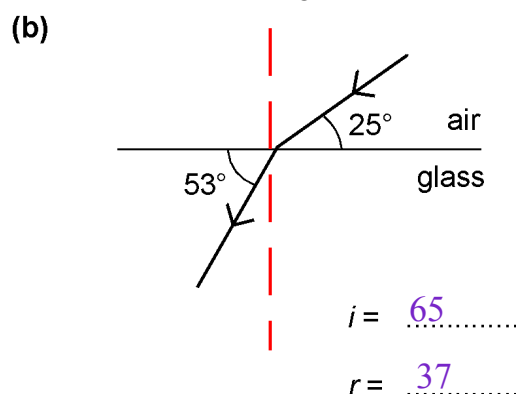
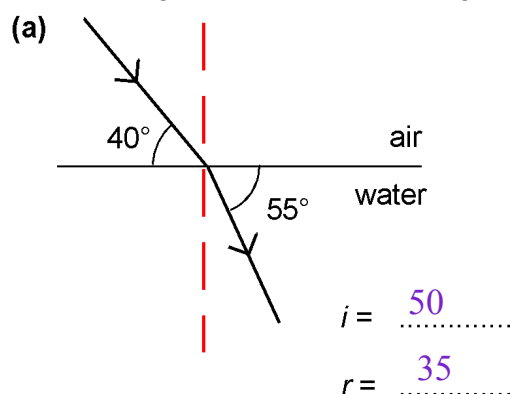
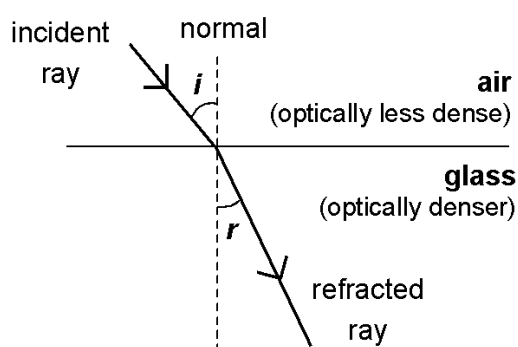
Type of medium			
Solid	Liquid	Gas	Vacuum
Slowest speed			
Densest (optical density)			



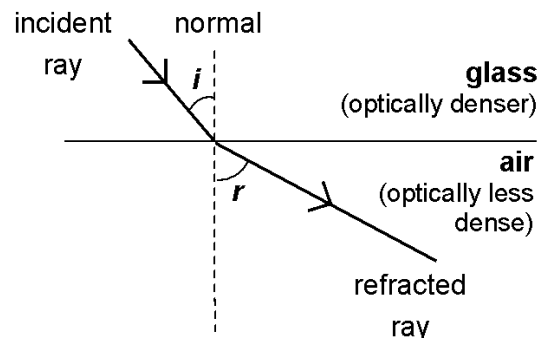
- **Normal** is an imaginary straight line that is perpendicular (90°) to the surface of the medium.
- **Angle of incidence** (i) is the angle between the incident ray and the normal.
- **Angle of refraction** (r) is the angle between the refracted ray and the normal.

Example 5

Determine the angle of incidence and angle of refraction for the followings.

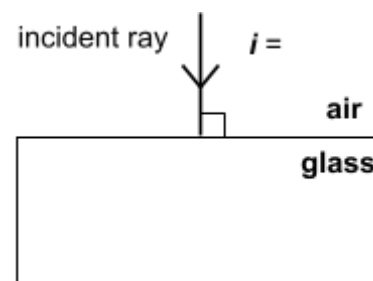
**Optically Less Dense $\rightarrow$ Denser**

light ray **bends towards the normal** when it enters an **optically denser medium** at an angle.

Optically Denser $\rightarrow$ Less Dense

light ray **bends away from the normal** when it enters an **optically less dense medium** at an angle.

When the **incident ray** is **perpendicular** to the surface of the glass block ($i = 0$), the ray of light is **not deflected**.



Do you know why the law of refraction does not apply when the incident ray is perpendicular to the surface of the glass block?

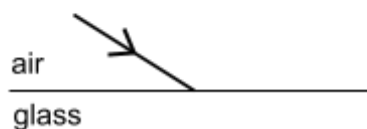
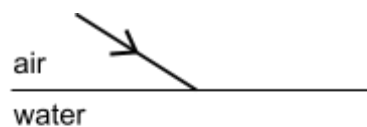
<https://www.youtube.com/watch?v=NLmpNM0sgYk>

(Fermat's Principle, Huygen's Principle and Life Guard Analogy does **not** explain it.)

Refractive Index (n)

- Each medium has a fixed value of refractive index.
- The refractive index indicates the **amount of bending of light** between two media.

medium	refractive index, n
water	1.33
plastic (perspex)	1.49
glass	1.52
diamond	2.42



- The **greater** the value of the refractive index
 - the **greater** the **bending effect of light**.
 - the **smaller** the **speed of the light in that medium**.
- The refractive index of a medium is

$\text{refractive index, } n = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} =$

Question:

Why does the speed of light slows down when it enters a glass block?

Isn't the speed of light is supposed to be a constant?

<https://www.youtube.com/watch?v=CiHN0ZWE5bk>

Example 6

- (a) State the speed of blue light travelling in vacuum.
- (b) Blue light travels at 2.1×10^8 m/s in glass.

Calculate the refractive index of the glass for blue light.

refractive index =

Example 7

Fig. 7.1 shows refraction of blue light at the boundary between air and glass.

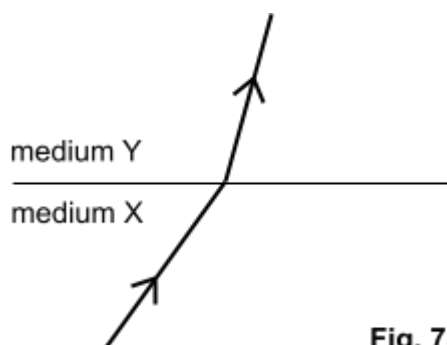


Fig. 7.1 (NOT drawn to scale)

- (a) State which medium, X or Y, represents glass.
Explain your answer.

Medium Y represents glass, X represents air. The OPTICALLY denser medium will bend the light ray towards the normal when the ray moves across 2 mediums`

- (b) Medium Y is replaced with medium Z. The refraction of the blue light at the boundary of medium X and Z is shown in Fig. 7.2.

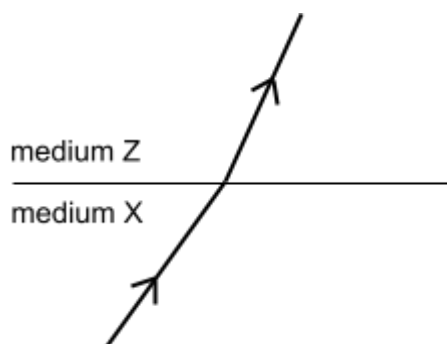


Fig. 7.2 (NOT drawn to scale)

- (i) Rank the medium X, Y and Z from the highest to the lowest refractive index.

Y, Z, X

- (ii) Explain your answer in (i).

From (a), medium Y is optically denser than X. From (b),
medium Z does not bend the ray as much as Medium Y.

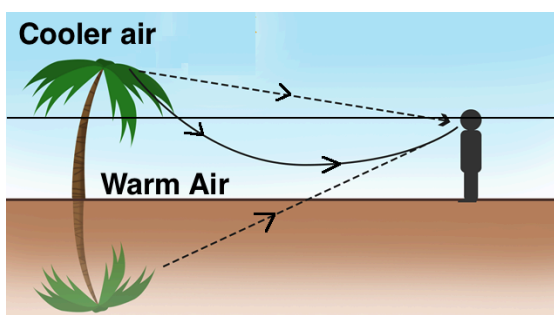
Effects of Refraction

Mirages & Fata Morgana

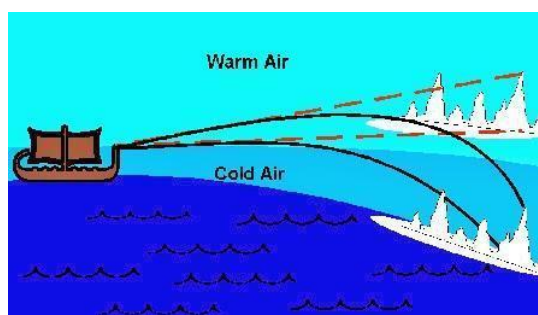
A traveler has lost his way in the desert. Enduring thirst and hunger, he suddenly saw an oasis, so the overjoyed man quickly ran towards it. To his great disappointment, it was just an illusion produced by a mirage. Such an episode was often pictured in movies, yet the optical magic that the nature plays with us - mirage - really exists in reality. Its formation is a result of the refraction and the total internal reflection of light in the air. Refer to the video here: <https://www.youtube.com/watch?v=mlAKYoV6AIU>.

To investigate the formation of a mirage, we firstly need to understand why light is refracted in the air. Regions of air at different temperatures have different refractive indexes, just like many different mediums. The closer the air is to the ground, the hotter it will be, and its refractive index will be smaller. We could imagine the air as many layers of medium with a particular refractive index for every layer, as shown in the diagram below. The surface, being warmer due to absorbing the heat of the sun, will have the lowest density. The layers above the surface will have a higher density. Any trees, which are originally higher up in the sky, will now appear on the ground due to the refraction of light.

Similarly, the city scapes will appear high up in the sky when the opposite happens. This is called Fata Morgana.



Mirage



Fata Morgana

Can you explain using diagrams, how Mirages and Fata Morganas are formed?

.....

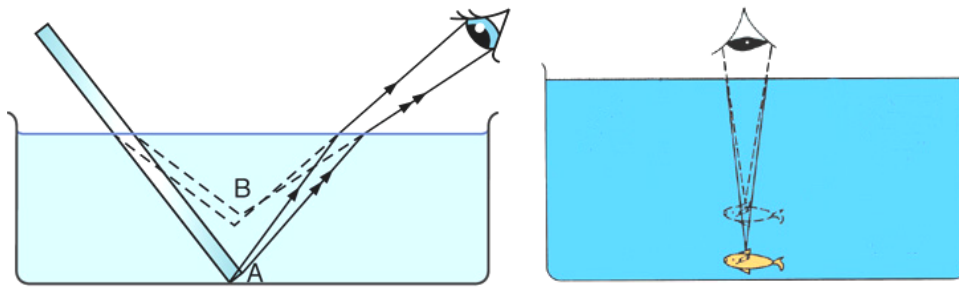
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Real depth and apparent depth

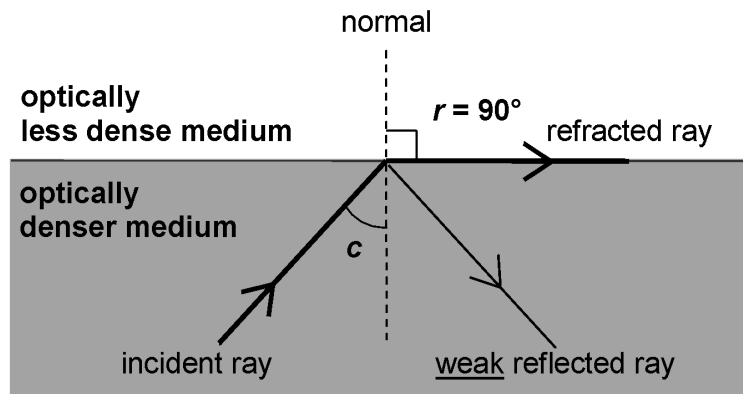
- The **apparent depth** is **less** than the **real depth**.
- Refraction causes point A to appear **nearer** to the surface at B.
- To the eyes, the straw appears to bend towards the surface of the water.



- To the eyes, the fish appears **nearer** to the surface of the water.

4.4 Total Internal Reflection: Critical angle (c)

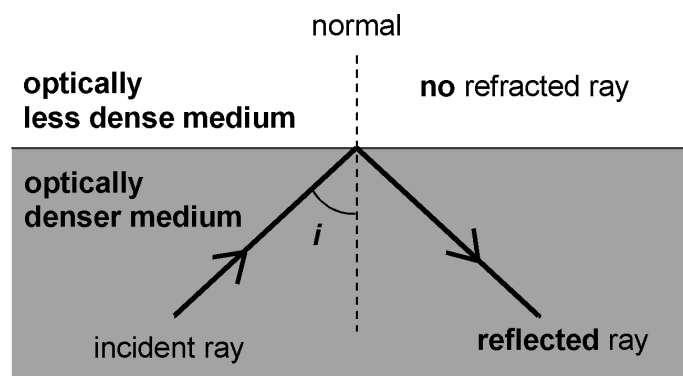
- Another effect of refraction is critical angle and total internal reflection. In situation where incident light rays travel from an optically denser medium to a less dense medium, it can be reflected when the ray enters at an angle beyond the critical angle.
- Critical angle (c) is **the angle of incidence in the optically denser medium for which the angle of refraction (r) in the optically less dense medium is 90°**



- It can **only** take place when light travels from an **optically denser** medium to an **optically less dense medium**.

Examples:

- from glass to air
 - from water to air
 - from glass to water
- When **total internal reflection** occurs, an **incident ray** is **reflected back in the same medium**
 - Hence, **no refracted ray is observed**



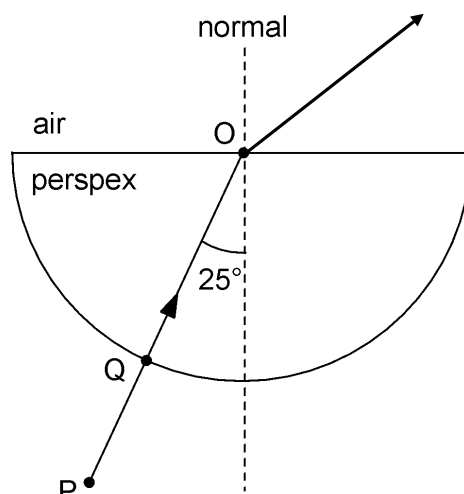
- Total internal reflection will only occur if **two conditions** are met.
- Light travels from **an optically denser medium strikes its boundary with an optically**

less dense medium

2. The angle of incidence (i) is greater..... than the critical angle (c).

Example 8

The diagram shows light entering a semi-circular Perspex block.



The ray of light PQ hits the Perspex block at Q.

O is the centre of the straight side of the block.

- (a) Explain why there is no change of direction of ray PQ as it passes into the perspex.

The ray is perpendicular to the surface at Q

- (b) The critical angle of the light in the Perspex block is 42° .

- (i) The ray of light QO makes an angle of 25° with the normal.
Complete the diagram to show the refracted ray.
- (ii) What would happen if the ray of light QO makes an angle of 48° with the normal? Explain your answer.

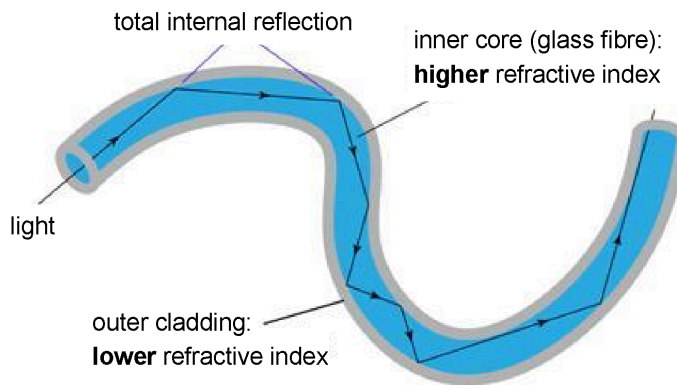
Total internal reflection will occur. The angle of reflection will be 48°

The ray moves from an optically denser to less dense material.

and that the incident angle is greater than the critical angle.

Application of Total Internal Reflection – Optical fibres

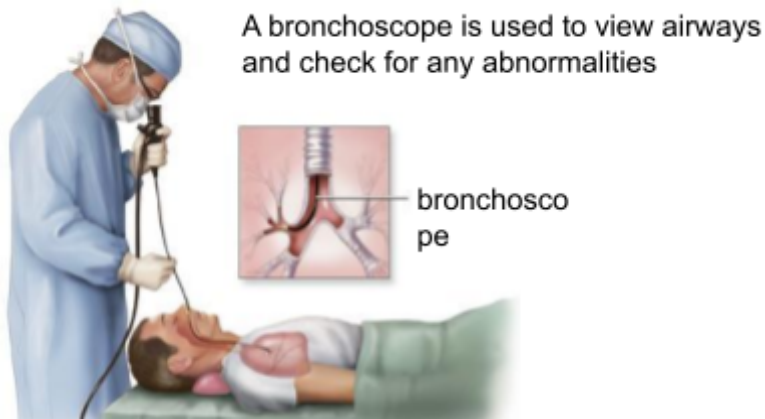
- **Optical fibres** are made of very thin glass fibres.
- It is used to transmit light or infrared radiation.



- To prevent the light rays from escaping from the sides of the glass fibre, the fibre is coated with a **cladding**.
- The cladding material has a **lower refractive index** than the fibre so that total internal reflection may occur at the fibre-cladding boundary.
- Each light ray reaches the boundary of the fibre with an **angle of incidence** *greater* than the **critical angle**.
- Each light ray in the fibre is totally internally reflected each time it reaches the boundary.
- Hence, each light ray entering a fibre at one end will leave the fibre at the other end even when the fibre bends around obstacles.

Applications

- Used in medicine to see the inside of the body without cutting the body open.



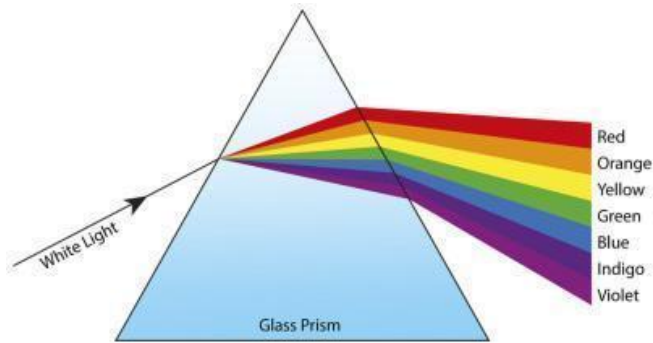
- Used in telecommunication to send signals securely.

Advantages of Using Optical Fibres in Telecommunication

Carry out research about why optic fibres are preferred over copper wires for internet transmission.

- Optical fibres are able to more information
than copper wires.
- Optical fibres experience less signal loss
than copper wires.
- Optical fibres can transmit information at a faster rate
than copper wires.

4.5 Dispersion



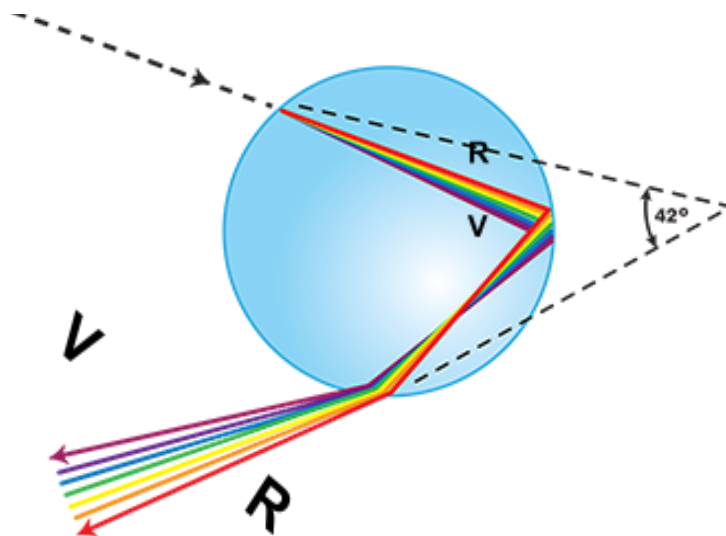
Dispersion

..... of light is the separation of light into its component colours.

- When a beam of white light passes through a glass prism, the light disperses and splits into a **range of colours** (spectrum).
- The prism refracts each colour by a **different amount**.

colour	violet	indigo	blue	green	yellow	orange	red
speed in glass	slowest						fastest
refraction	Bend the most						Bend the least

A rainbow is formed when light enters each water droplet, and the different colours bend (refract) at slightly different angles. They reflect off the inside of the drop before refracting again as they leave. The shorter wavelengths refract more.

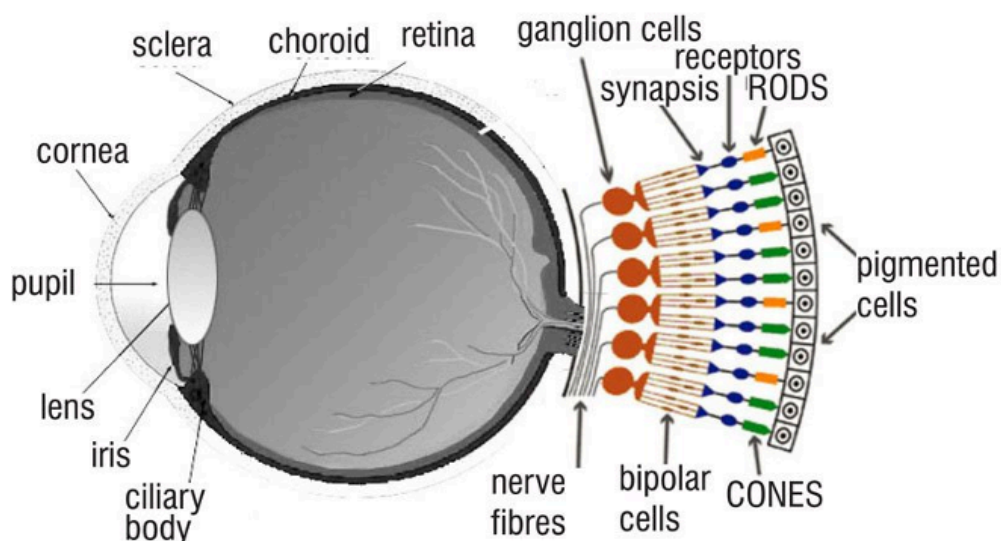


The simple rainbow is formed through the refraction, total internal reflection and refraction of a spherical water droplet in the air. This will cause the white light to split up into its individual components and form a rainbow in the sky. The next time you see a

rainbow, try to apply your concepts of dispersion of white light.

4.6 Seeing colours

The structure of the eye is as shown in the diagram below:



A team of SST students participated in a video making competition and here is an interesting presentation of how we see colours:

https://www.youtube.com/watch?v=_slhdOT9VIQ

Do you have colour deficiency?

There are many colour blindness tests on the internet:

<https://www.colorlitelens.com/color-blindness-test.html>

Go here for the Ishihara test: <https://www.colorlitelens.com/ishihara-test>

Other tests: <https://www.colorlitelens.com/color-vision-test.html>

Our eyes have different levels of Red, Green and Blue cones. As such, we see colours differently.

What is the white LED light?

White LED light consists of individual red, blue and green components, so it doesn't have a single wavelength or frequency. White LED devices have a mixture of the frequencies 474 THz, 535 THz and 638 THz. The development of white LED devices has led to cheap, energy-efficient lighting that can be used in a variety of settings, from streetlamps to desk lights.

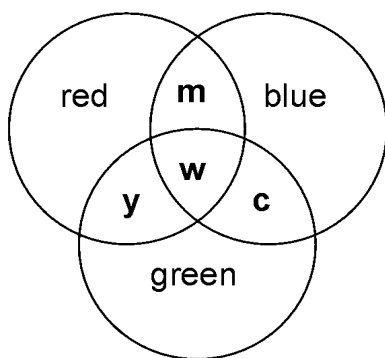
Frequency / THz	Frequency / Hz	Speed / m/s	Wavelength / m	Wavelength / nm	Colour
474	4.74×10^{14}	3.0×10^8	6.33×10^{-7}	633	RED
535	5.35×10^{14}	3.0×10^8	5.61×10^{-7}	561	GREEN
638	6.38×10^{14}	3.0×10^8	4.70×10^{-7}	470	BLUE

Mixing Colored Light

State the colour of the LED light if some or all of the 3 frequencies are switched on.
1 represents the ON state and 0 represents the OFF state.

Combination	RED	GREEN	BLUE	OBSERVED
1	1	1	1	WHITE
2	1	1	0	YELLOW
3	1	0	1	MAGENTA
4	1	0	0	RED
5	0	1	1	CYAN
6	0	1	0	GREEN
7	0	0	1	BLUE
8	0	0	0	OFF

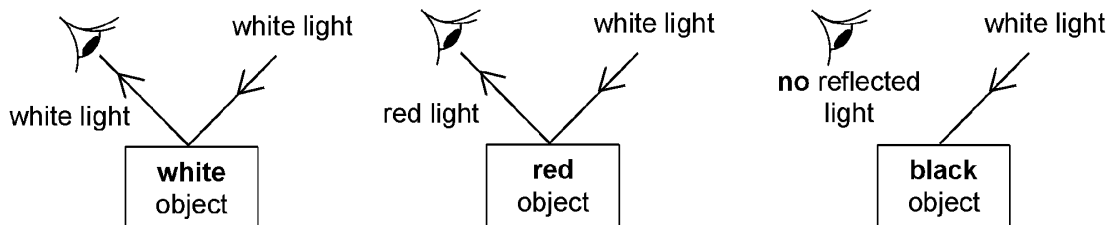
The venn diagram below shows what happens when the Red, Green and Blue lights are superimposed on each other. Indicate the colours y, m, c and w below.



y: yellow
m: magenta
c: cyan
w: white

Seeing colored objects in white light

- When white light shines on a colored object, some of the colors in the spectrum are reflected and some are absorbed.
- The color of an object is the color of the light that is reflected from it into our eyes.

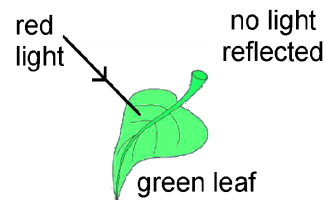
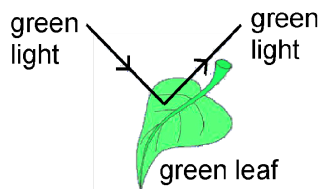


- | | | |
|---|---|---|
| <ul style="list-style-type: none"> White objects appear white as they reflect all | <ul style="list-style-type: none"> Red objects reflect red light and absorb all the other colours. | <ul style="list-style-type: none"> Black objects absorb all the light that falls |
|---|---|---|

colours and absorb none .		on it and does not reflect any light.
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- **Black** is **not** a colour of light. It indicates the total absence of reflected light.

Seeing coloured objects in coloured light



▪ In green light, a green leaf appears green because it reflects green light.	▪ In red light, a green leaf appears black because it absorbs the red light and does not reflect any light.
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4.7 Effects of light

Advantages of light produced by technological advances in LED lighting

- Night lighting along streets that improve safety
<https://science.howstuffworks.com/environmental/green-tech/sustainable/led-streetlight1.htm>
- Large LED TV screens is now possible
<https://www.cnet.com/news/samsung-microled-makes-massive-modular-tv-a-reality/>

Adverse effect of light on our society and the environment

- Light pollution:
<http://www.darkskiesawareness.org/faq-what-is-lp.php>
- Disorientation of birds
<https://www.nytimes.com/2018/09/10/opinion/9-11-tribute-in-light-birds.html>
- Waste of energy <https://www.darksky.org/light-pollution/energy-waste/>

THE END